AL-RIYADA COLLEGE FOR HEALTH SCIENCES · MEDICAL HRMS

Installation and Setup Guide

Install Medical HRMS and the Jameah (Ministry) data views on a Frappe server, set them up for the college, and check that everything works.

FOR	The server administrator who installs and updates the system, and the system owner who sets it up (HR Manager / System Manager).
COVERS	Medical HRMS app (`medical_hrms`) including the Jameah views, branch `develop` .
VERSION	27 October 2026 (Round 2)
BUILT BY	Abdul Hakeem (Medical HRMS and Jameah lead)
SEE ALSO	User Guide (how HR, managers, employees and testers use the system).

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1 What gets installed

Medical HRMS is one Frappe app, `medical_hrms`. It adds the college's HR processes on top of the standard Frappe HR (HRMS) app, and the Jameah views the Ministry's data agent reads. The standard apps are not modified.

Part	What it gives the college
HR Portal	One page for HR: requests waiting for a decision, leave setup check and sick leave, recruitment, Ministry data, academic presence (`/app/hr-dashboard`).
Employee Portal	Self-service for every employee: 11 request types, the leave flow (plan, apply, return), balances and monthly limits (`/app/employee-dashboard`).
Requests	Permission, Remote Work, Salary Certificate, Overtime pre-approval, Training, Children Medical Allowance, Company Car, Employee Data Update, Leave Plan, Leave Return, Contract Renewal (HR only), Medical Licence.
Leave	Editable starter leave types and policies (annual 21 / 30 days), sick leave in 3 pay stages with its own sick year, monthly half-day and work-from-home limits.
Recruitment	Job Application Links, the public application form, review, and Create Employee (with login and, for academic roles, Instructor / Academic Advisor / Head of Department records).
Ministry codes	9,867 official Ministry lookup codes, loaded on every migrate, used on Employee and applicant fields.
Jameah views	23 read-only database views in the Ministry's format: Employees 5, Faculty 8, Students 10, with a Data Check page, a Code Map and an independent verification.
Academic presence	Faculty and proctor schedules, students on campus per hour, faculty after 4 PM.

2 Before you start

- A Frappe **v15** bench with these apps installed on the site: `frappe`, `erpnext`, `hrms`, `education`.
- MariaDB, Redis and Node (as for any Frappe bench). Medical HRMS needs no extra packages.
- Workers running for the **short, default and long** queues, and the **scheduler** enabled. The Jameah academic figures are rebuilt every hour on the long queue.
- Access to the GitHub repository `phpdevpro/edu-alriyada-hrm`. The branch to install is `develop`.
- A backup of the site before you install or update: `bench --site <site> backup --with-files`.

> **The education app must be current**
>
> The Ministry GPA and credit hours use the college's official calculation that comes with the latest `education` app (master branch). Update it on the server in the same deployment.

3 Install on a new site

3.1 Install the app

Run these commands as the bench user, in the bench folder. Replace `<site>` with the site name.

```
cd ~/frappe-bench
bench get-app git@github.com:phpdevpro/edu-alriyada-hrm.git --branch develop
bench --site <site> install-app medical_hrms
bench --site <site> migrate
bench build --app medical_hrms
bench --site <site> clear-cache
bench restart
```

The app folder must end up as `apps/medical_hrms`, and `sites/apps.txt` must list each app on its own line (see Troubleshooting).

> **Read the migrate output**
> It must finish without a `Traceback`. The last part rebuilds the Jameah academic figures and takes a few minutes. If it prints a line starting with *Kept (they hold data ...)*, send that line to the developer.

3.2 What migrate sets up

Every `migrate` runs the app's setup steps, so a new site and an updated site end up the same. Nothing has to be run by hand.

Step	What it does
Ministry codes	Loads or updates the official Ministry lookup codes (Jameah Ministry Code).
Ministry fields	Adds the Ministry and Jameah fields and tables to Employee and Instructor (identity, nationality, job rank, qualifications, experience, ...).
Jameah	Creates the Jameah doctypes and settings, refreshes the Code Map, creates the 23 views and rebuilds the academic figures.
HR permissions	Gives HR roles the access their pages need (for example read on Company, Holiday List, Instructor).
Employee Self Service	Creates the employees' workspace.
Leave starter records	Creates the leave types (annual, Sick Leave 1 / 2 / 3, unpaid) and draft policies if they are missing. Existing records and HR's changes are kept.
Monthly limits, presence	Adds the HR Settings fields for monthly half-day / work-from-home limits and for academic presence.
Requests	Leave Application gets the *Rejection Reason* and *Sent Again From* fields (the other requests have them built in), and its status changes are kept so employees see the history. The two leave e-mail templates are created and chosen in HR Settings where none is chosen (section 6.2).

Step	What it does
Instructor / Academic Advisor	Stops these forms copying the Employee's gender and department into their own lists (which hold different values); the matching values are filled in instead.
Patches	One-time data changes listed in `patches.txt` (for example the sick-leave renaming).

> **Do not run the old setup scripts**
>
> Earlier notes listed scripts such as `configure_hrm_request_governance`, `quick_grant_hr_permissions` and `apply_hr_workspace_layout`. Do not run them: the doctypes now carry their own permissions, and `configure_hrm_request_governance` would replace them.

4 Update an existing server

1. Back up the site: `bench --site <site> backup --with-files`.
2. Get the new code:

```
cd ~/frappe-bench/apps/medical_hrms
git fetch origin
git checkout develop
git pull --ff-only origin develop
```

3. Update the `education` app to its current master in the same deployment (see section 2).
4. Apply it:

```
cd ~/frappe-bench
bench --site <site> migrate
bench build --app medical_hrms
bench --site <site> clear-cache
bench restart
```

5. Run the checks in section 5.

> **Timing**
>
> The hourly Jameah refresh makes pages slower for a few minutes while it runs. A migrate never fails because of it any more (only one refresh runs at a time), but schedule updates outside busy hours.

5 Check the installation

Check	How	Expected
Scheduler and workers	`bench --site <site> doctor`	Scheduler enabled; workers for short, default and long.
Ministry views	`bench --site <site> execute medical_hrms.jameah.verify.run`	Every view **OK** (23).
Error Log	Open `/app/error-log`, newest first.	No new errors since the update.
HR login	Log in as an HR Manager.	The HR Portal opens.
Employee login	Log in as an employee whose user is linked to an Employee record.	The Employee Portal opens with their name.
System e-mail	Email Account *Notifications* → send a test e-mail.	The e-mail arrives.
Notifications	As an employee, send a request. As an HR user, click the bell on the HR Portal.	"New <request> from <employee>" within a few seconds (made by a background worker).
Logo	Open `/files/rchs-logo.png`.	The college logo shows.

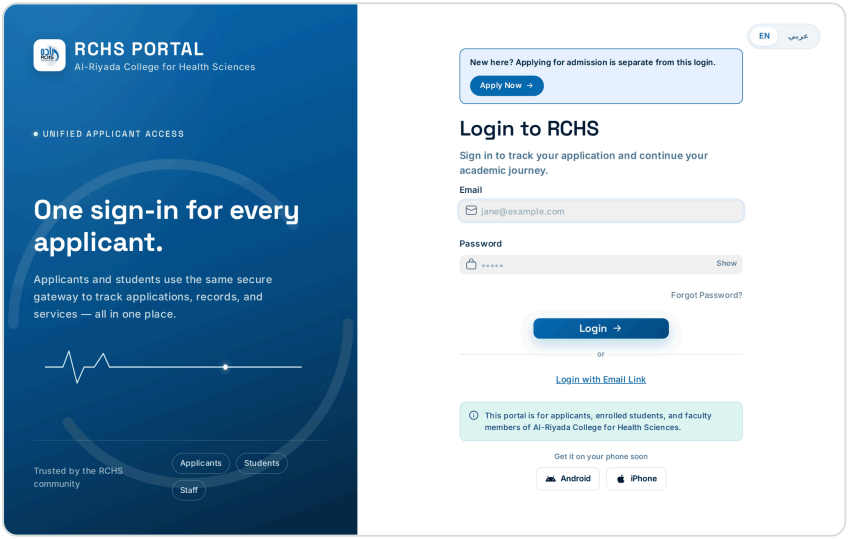


Figure 1. The login page. HR users land on the HR Portal and employees on the Employee Portal.

6 First-time setup

Done once by the system owner (HR Manager or System Manager) after the first installation, and checked at the start of every leave year.

6.1 Users and roles

Every person who logs in needs a **User**. Employees also need their User linked on their **Employee** record (field **User ID**); this gives them the Employee role and the Employee Portal.

Role	Who has it	What it allows
HR Manager	Head of HR, HR system owner	Everything in the HR Portal, including the staff data sheets, starting a sick leave year, Jameah Settings and the Code Map.
HR User	HR officers	Process requests, employees, leave and recruitment. Read-only on Jameah Settings and the Code Map; cannot use the staff data sheets or start a sick leave year.
Employee	Every member of staff	The Employee Portal: own requests, own leave, own records only.
Leave Approver	Managers named as leave approver	Approve the leave of the employees they approve for.
Department Head	Heads of department	Approve training, travel and expense claims above the department-head limit.
Finance Manager / Accounts Manager	Finance	The finance step on money requests and amounts above the finance limit.
Education Manager	Academic Affairs	Jameah Data Check, Jameah Settings and the Code Map.
System Manager	IT	Everything (administration).

Approval limits by amount (checked when a request is approved):

Request	Department Head	HR Manager	Finance
Training	above 10,000	above 30,000	above 50,000
Travel	above 15,000	—	above 50,000
Expense claim	above 5,000	—	above 20,000
Company car	—	—	always (final approval)
Children medical allowance	at most 3 children per academic year; at most 30,000 approved per child		

6.2 Company and e-mail

- Check the **Company** record of the college (Al Riyada College): default holiday list (section 6.3) and the company e-mail domain.

- Academic hires must use their official college e-mail (`@alriyada.edu.sa`) as Company Email; the Create Employee dialog asks for it.
- Open Email Account **Notifications**, enter its password again and save, then send a test e-mail. Until this works the system cannot send e-mails.
- Request notifications (new requests to HR, each later step to the employee) appear in the bell and are e-mailed through this account. Each user can switch the e-mails off in My Settings → Notifications.
- Leave e-mails: the migrate chooses *Leave Approval Notification (Medical HRMS)* and *Leave Status Notification (Medical HRMS)* in HR Settings → **Leave Approval Notification Template** and **Leave Status Notification Template**, only where they are empty, so a template HR chose is kept. The approver's e-mail lists the application; the employee's says approved, rejected (with the reason) or cancelled, and opens the application on the Employee Portal. HR can change the wording in Email Template. Without a template HRMS sends no leave e-mails.

6.3 Leave, in this order

The HR Portal has a **Leave** section with a **Leave setup check** per company and the yearly setup steps in order. Work through the steps until every line of the check is green.

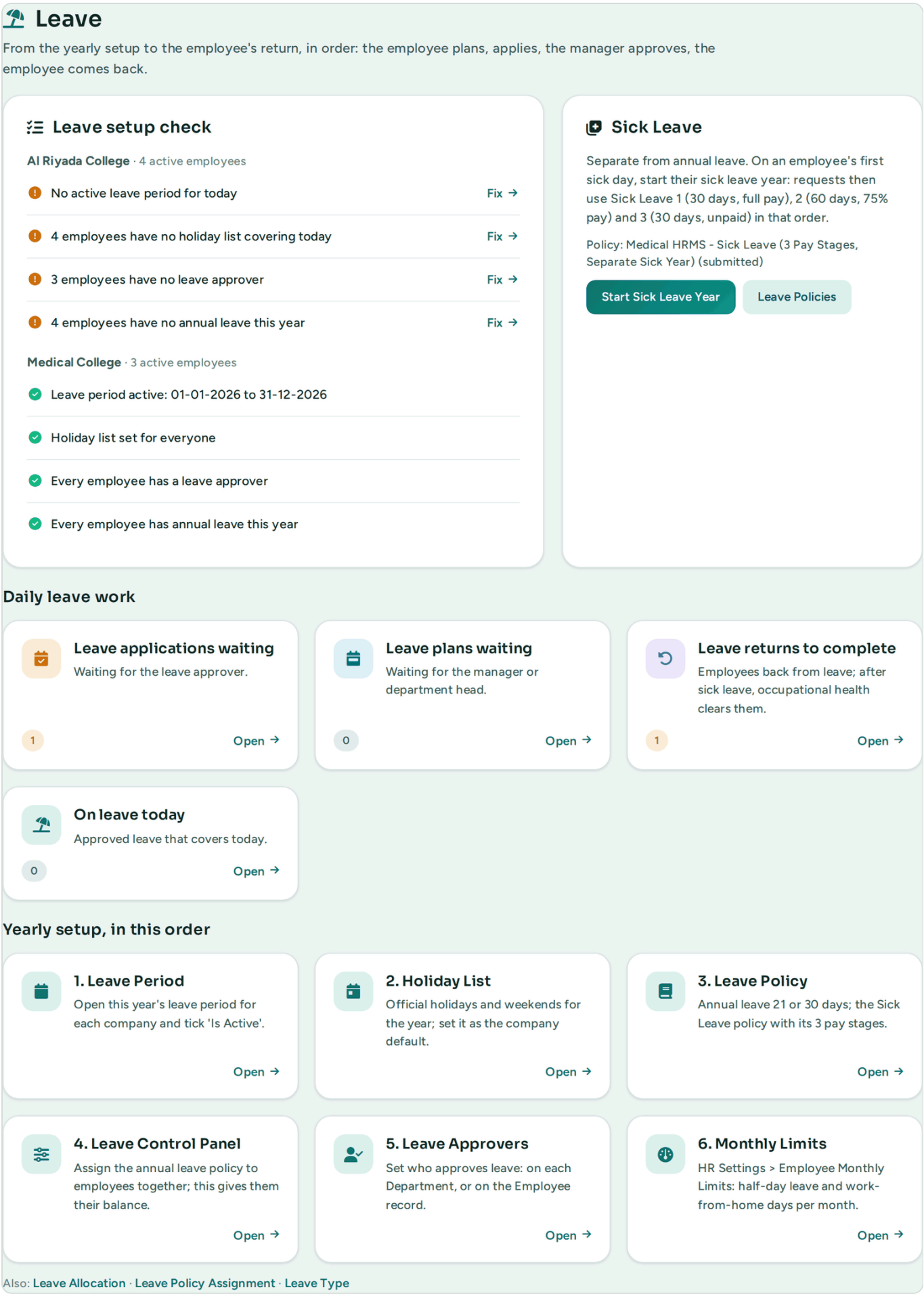


Figure 2. HR Portal → Leave. The setup check shows what is missing per company (orange, with a Fix link) and what is done (green). Below: daily leave work and the yearly setup steps.

1. **Leave Period**: create this year's leave period for the company and tick *Is Active*. Migrate creates an *inactive* starter period; check its dates and activate it.
2. **Holiday List**: enter the official holidays and weekends for the year and set it as the company's default holiday list.
3. **Leave Policy**: review the draft policies (annual leave 21 days, annual leave 30 days for five or more years of service, and the Sick Leave policy) and **submit** the ones the college uses.
4. **Leave Control Panel**: assign the annual leave policy to the employees together. This gives them their balance.
5. **Leave Approvers**: set who approves leave, on each Department (Leave Approvers table) or on each Employee.
6. **Monthly Limits**: HR Settings → Employee Monthly Limits (section 6.5).

> **Until this is done, employees cannot apply for leave**
> An employee with no annual leave assigned sees "No leave to apply for yet". Check the Leave setup check first when someone reports a leave problem.

6.4 Sick leave

Sick leave is separate from annual leave and follows the Saudi Labour Law: **Sick Leave 1** 30 days full pay, then **Sick Leave 2** 60 days at 75% pay, then **Sick Leave 3** 30 days unpaid, within one sick year that starts on the employee's first sick day.

1. Submit the policy *Medical HRMS - Sick Leave (3 Pay Stages, Separate Sick Year)*. Finance checks the 75% pay on a salary slip.
2. On an employee's first sick day, HR opens HR Portal → Leave → **Start Sick Leave Year**, picks the employee and the first sick day. This creates their Sick Leave 1 and 2 balances for one year; annual leave is not touched.
3. The employee then applies for sick leave from the Employee Portal; the stages are used in order and a request that crosses a stage must be split.

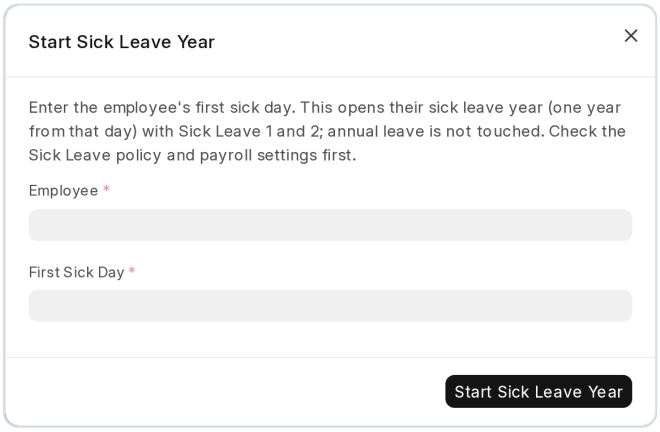


Figure 3. Start Sick Leave Year (HR Manager).

6.5 Monthly limits and academic presence

In **HR Settings**:

Field	Meaning
Enable Monthly Half-day Limit	Limit half-day leave per calendar month.
Leave Type Subject to Half-day Limit	Which leave type the half-day limit applies to (usually annual leave).
Half-day Requests per Calendar Month	The limit.
Enable Monthly Work-from-home Limit	Limit remote-work days per calendar month.
Work-from-home Working Days per Calendar Month	The limit.
Academic Holiday List	Holiday list used by Academic Presence (days without classes).
Suspends Regular Classes	Event types that cancel classes on that day.
Faculty After-hours Cut-off	The time used by "Faculty After 4 PM".

Data the Academic Presence pages need

Faculty Schedules and Students on Campus read the published class timetables and the attendance teachers record. Proctor Schedules reads the exam committees and their proctors, and is empty until they are entered. Faculty After 4 PM uses attendance-device check-ins when devices are connected (HRMS Employee Checkin); until then it uses the timetable and class attendance.

7 Jameah (Ministry) setup

Jameah is the Ministry's data system. Its agent reads the college's data from read-only database views in the Ministry's format. Medical HRMS creates and maintains these views; this section sets them up and checks them.

7.1 The 23 views

Group	Views	One row per
Employees (5)	jameahEmployeePersonalInfo, …AcademicQualifications, … AcademicExperience, …PreviousExperience, … ProfessionalCertificates	Non-teaching employee of the reported company; table row
Faculty (8)	jameahFacultyPersonalInfo, …Research, …AcademicQualification, …AcademicExperience, …PreviousExperience, … ProfessionalCertificates, …Awards, …Attachments	Active Instructor; table row
Students (10)	jameahStudentPersonalInfo, …Companion, …ForeignAddress, … Relatives, …KSAAddress, …HighSchoolCertificate, …Contact, … Scholarship, …AcademicDetails, …Attachments	Student; guardian / certificate / document row

- **Codes, not names**: lookup columns carry the official Ministry code. A value that is not a valid code for its list is sent empty, never guessed.
- Teaching staff are sent once, as **Faculty**, never also as Employees.
- Column names follow the Ministry document exactly (including its spellings `EnglishFourhtName` and `Summury`).

7.2 Jameah Settings

Open **HR Portal → Ministry (Jameah) → Jameah Settings**. Values left empty are sent empty. Academic Affairs confirms them with the Ministry.

Setting	Meaning
Educational Institution (UniversityId)	RCHS is **P18** in the Ministry list.
College (AdmissionCollegeId / CurrentCollegeId)	The college code the Ministry assigned to RCHS.
Country of Study	Country of the campus.
Study Location City	Campus city, used when a student has no study city.
Study Type for Regular / Bridging Students	Study type sent for regular and for bridging students.
GPA Type	The grading scale of the GPA column.
Program Duration and Unit	Standard duration of the programme.
File Base URL	Prefix for attachment links, e.g. `https://erp.alriyada.edu.sa`.
Company Reported as Employees	Only this company's (non-teaching) employees are sent.

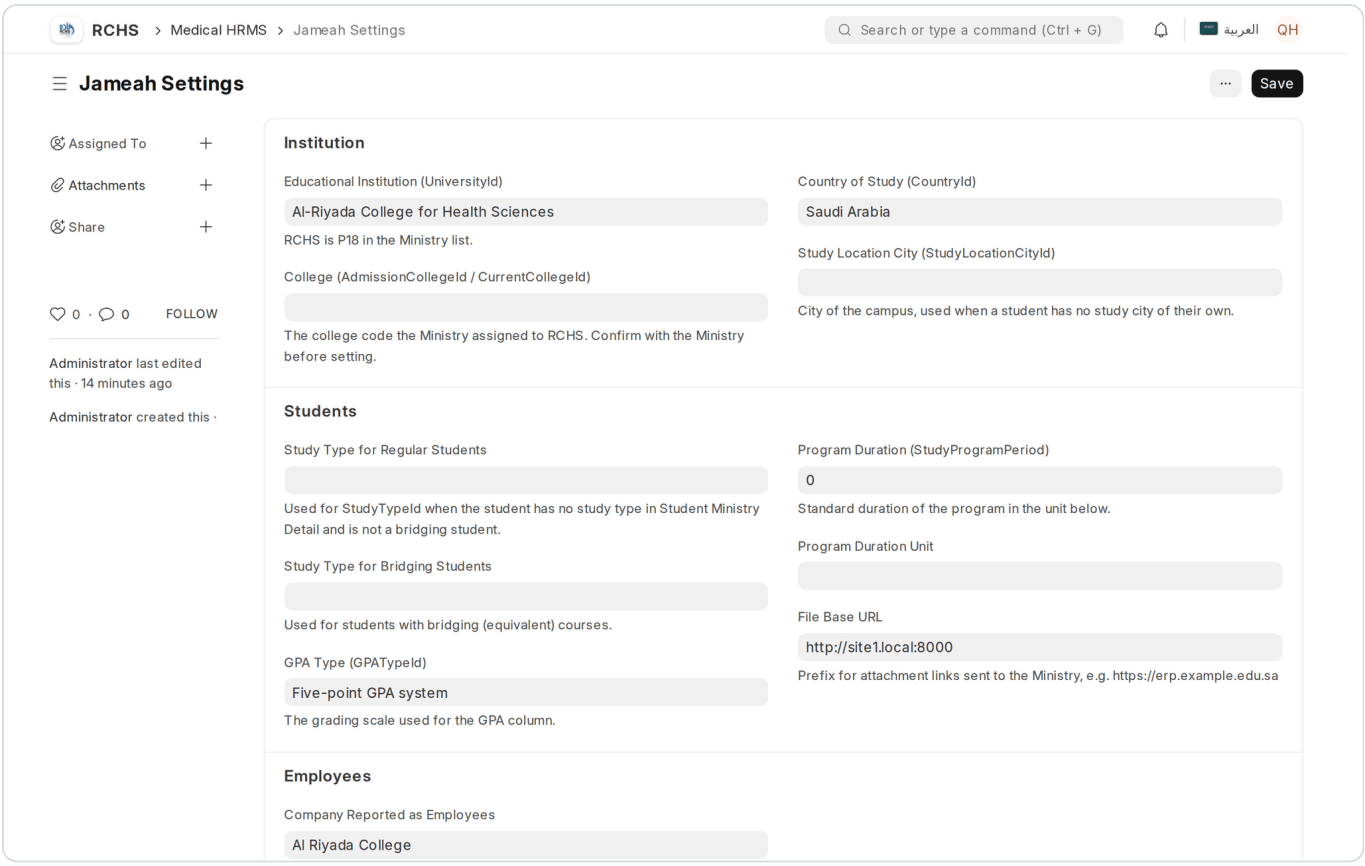


Figure 4. Jameah Settings.

7.3 Jameah Code Map

Some college lists keep their own codes (for example every city has code 90). The **Code Map** links each college value to the official Ministry code. Rows whose code and name agree with the official list are verified automatically; the others wait for a person.

1. Open **Ministry (Jameah) → Code Map**, filter *Status = Needs Review*.
2. Open a row, pick the Ministry code, set *Status = Verified*, save.
3. Until a row is verified, that value is sent empty.

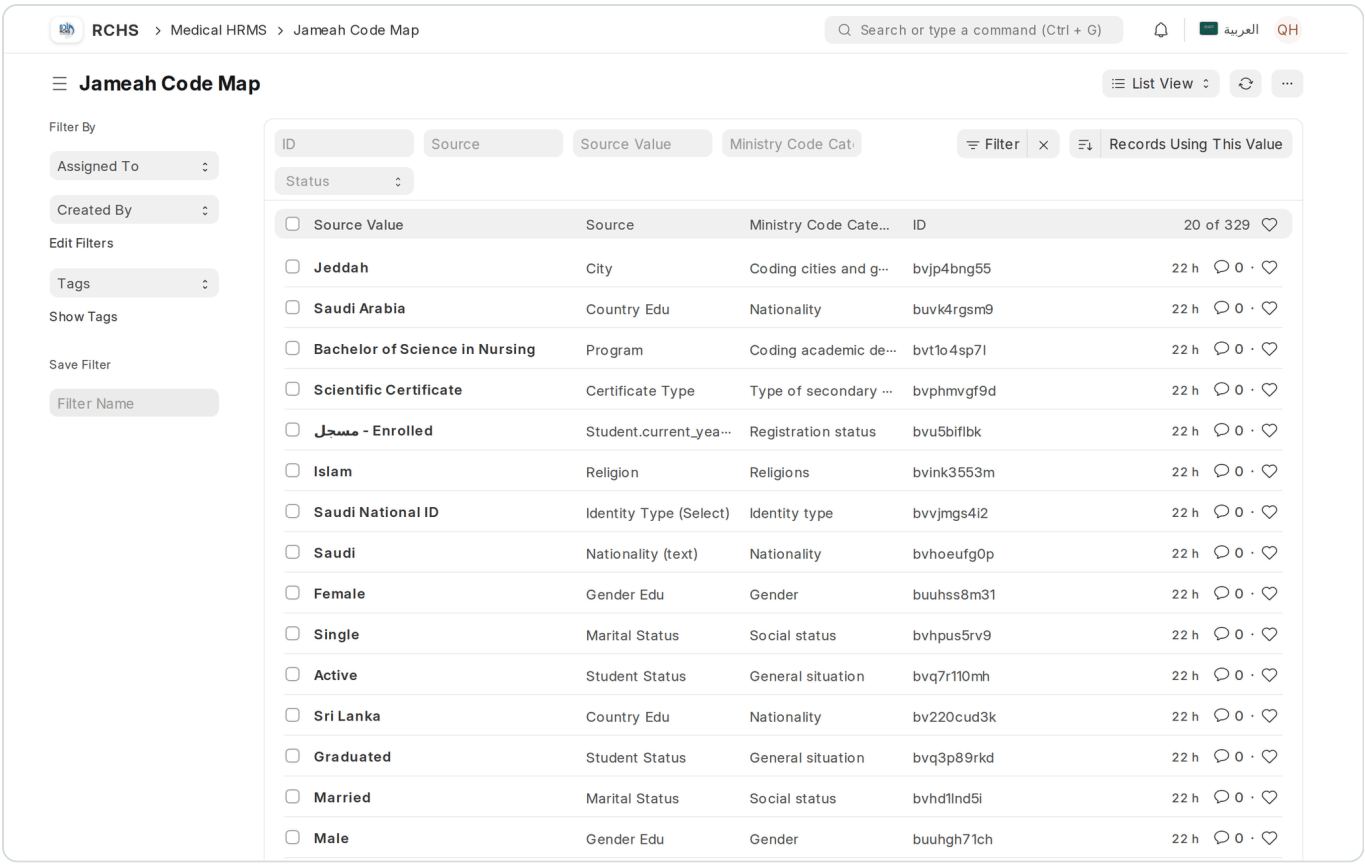


Figure 5. Jameah Code Map.

7.4 Jameah Data Check

The **Jameah Data Check** page shows what the views would send right now: invalid codes (must be 0), values waiting for a code, settings not filled, mandatory columns with gaps and problems in the source records. Use it after every data change.

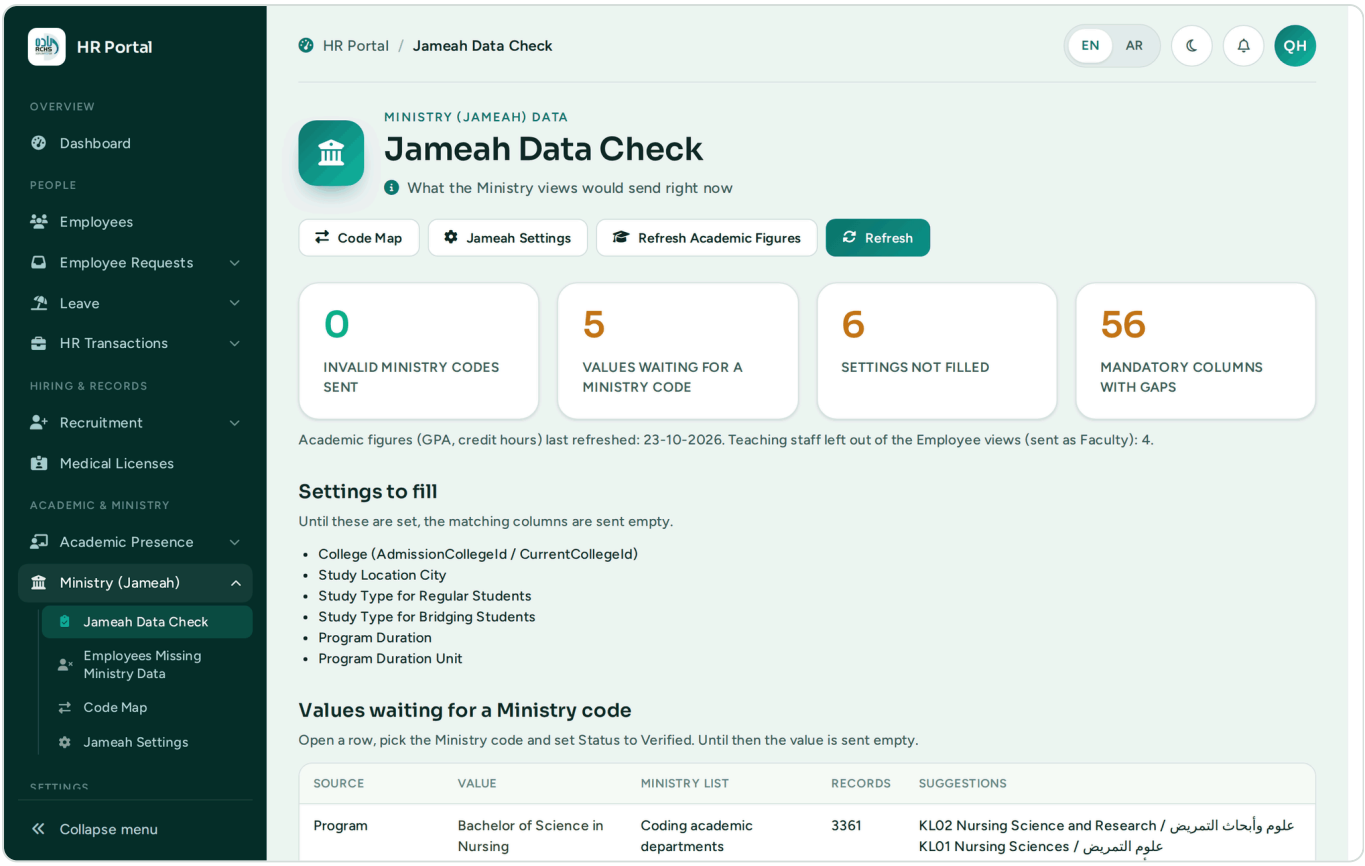


Figure 6. Jameah Data Check. Buttons at the top open the Code Map and Settings and refresh the academic figures.

7.5 Academic figures

GPA, credit hours and the latest enrolment come from a summary table rebuilt every hour (long queue, about 10 minutes), after every migrate, and with **Refresh Academic Figures** on the Data Check page. They equal the transcript (the college's official calculation). Only one rebuild runs at a time.

7.6 Verify every view

```
bench --site <site> execute medical_hrms.jameah.verify.run
```

Rebuilds every view independently in Python and compares it with the database view value by value. Every line must say **OK**; it also lists mandatory gaps and invalid codes per view.

7.7 Account for the Ministry agent

```
bench --site <site> execute medical_hrms.jameah.views.grant_statements
```

Prints the `CREATE USER` and `GRANT SELECT` statements for a read-only account that can read the 23 views and nothing else. Run them as the MariaDB administrator with a strong password, then give the Ministry the host, port, database, user and password.

7.8 Completing staff data

Non-teaching staff (HR Manager): open **Ministry (Jameah) → Employees Missing Ministry Data**. The report lists every employee the views send with each Ministry-mandatory value still empty.

1. **Download Data Sheet**: an Excel sheet with the current values; empty mandatory cells are red and Ministry lists are dropdowns.
2. HR fills it in.
3. **Check / Import Data Sheet → Check (nothing is saved)**: a result sheet lists each row's status.
4. When the check is clean, **Import**: values are saved through the normal Employee checks.

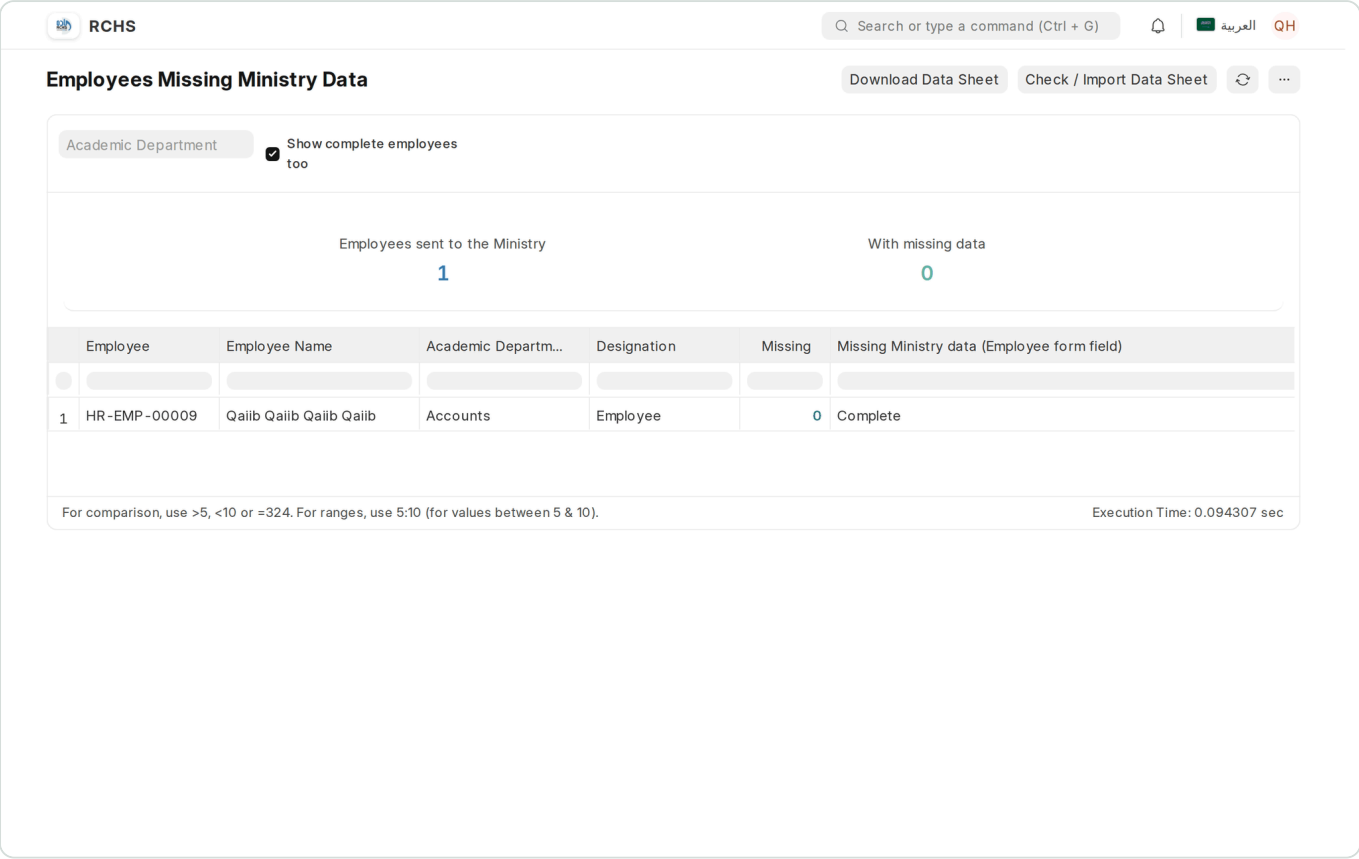


Figure 7. Employees Missing Ministry Data, with the data sheet buttons.

Teaching staff (existing Instructors without an Employee record): the administrator exports a sheet for HR, HR fills it, then it is checked and imported:

```
bench --site <site> execute medical_hrms.instructor_data_sheet.export_sheet
# HR fills the sheet and uploads it (Desk > File); check it (nothing is saved):
bench --site <site> execute medical_hrms.instructor_data_sheet.import_sheet --kwargs "{'file':
'/private/files/<name>.xlsx'}"
# when the check is clean, import for real:
bench --site <site> execute medical_hrms.instructor_data_sheet.import_sheet --kwargs "{'file':
'/private/files/<name>.xlsx', 'commit': 1}"
```

8 Testing the installation

> **Test sites only**
> The automatic browser tests create test logins with known passwords. They refuse to run unless the site is named in `QA_TEST_SITE`. Never run them on the live server.

Test	Command (from `apps/medical_hrms`)	Passes when
Every file	`python3 tests/checks/files.py`	PROBLEMS: 0
Every hook, doctype and record	`../../env/bin/python tests/e2e/fx.py tests/checks/site.py`	PROBLEMS: 0
Every server method, by role	`python3 tests/checks/api_methods.py` (bench running, QA logins loaded)	TO REVIEW: 0
Unit tests	`bench --site <test site> run-tests --app medical_hrms` (once per test site: `bench --site <test site> set-config allow_tests true`)	OK
Ministry views	`bench --site <site> execute medical_hrms.jameah.verify.run`	Every view OK
Browser suite (95 checks)	See `tests/e2e/README.md`: load the QA data, run `e2e_full.py`, remove the QA data	95/95 passed

Results of these checks go into the **Test Cases** workbook in `project_tracking/` (column *Last automated run*). The testing team then works through the same workbook by hand; see the User Guide, section "Testing the system".

> **Start the browser suite just after the hourly Jameah refresh**
> While the refresh runs, pages load slowly and checks can time out.

9 Troubleshooting

Problem	What to do
"App medical_hrms not in apps.txt"	Put each app on its own line in `sites/apps.txt` (frappe, erpnext, payments, education, hrms, medical_hrms), then install and migrate again.
An employee sees "No leave to apply for yet"	They have no annual leave assigned: HR Portal → Leave setup check (section 6.3).
"Failed to decrypt key ... password" in the Error Log	Email Account *Notifications*: enter the password again and save (section 6.2).
Pages slow for a few minutes every hour	The Jameah academic summary is being rebuilt. Check the Scheduled Job Log (*summary.refresh*). On the live server, report it if it is noticeable.
An HR user without the Employee role gets "Not permitted" on Employee Self Service	Expected: that workspace is for employees. HR works from the HR Portal.
Academic hire refused with "College E-mail Needed"	Enter the official `@alriyada.edu.sa` address as Company Email in the Create Employee dialog.
A Ministry column is empty	Jameah Data Check: a setting not filled, a Code Map row not verified, or data missing on the record (report / data sheet).
Logo missing on every page	Upload `rchs-logo.png` again (Website Settings / File).
No notification in the bell after a request is sent	Notifications are made by a background worker (short queue) just after the save: check the workers with `bench doctor`, and that the user is enabled and has notifications on (My Settings → Notifications). With a single worker for every queue (a development copy), they wait while the hourly Jameah refresh runs (up to about 15 minutes); production keeps separate short, default and long workers.
"Please set default template for Leave Status Notification"	HR Settings has no leave e-mail templates: run the migrate again, or choose the two *(Medical HRMS)* templates there (section 6.2).
An instructor or head of department sees no My HR card on their dashboard	Their login is not linked to an Employee record (the link also gives the Employee role): link it on the Employee (User ID) or with the Instructor data sheet. If it is linked, run `bench build --app medical_hrms` and `clear-cache`, then reload the page. The card is on the Instructor, Department Head, Dean, Vice Dean, IT, Academic Operations, Student Affairs, Admission, Finance, Clinical and Advisor dashboards.

10 Command reference

```
# install (first time)
bench get-app git@github.com:phpdevpro/edu-alriyada-hrm.git --branch develop
bench --site <site> install-app medical_hrms

# every update
cd apps/medical_hrms && git fetch origin && git checkout develop && git pull --ff-only origin
develop && cd ../..
bench --site <site> migrate
bench build --app medical_hrms
bench --site <site> clear-cache && bench restart

# checks
bench --site <site> doctor
bench --site <site> execute medical_hrms.jameah.verify.run

# Jameah
bench --site <site> execute medical_hrms.jameah.views.grant_statements      # read-only account
for the Ministry

# Ministry codes from a new Ministry workbook
bench --site <site> execute medical_hrms.import_ministry_excel_to_jameah_codes.execute --kwargs
"{'file_path':'/absolute/path/Ministry Code - (Translated).xlsx'}"

# teaching-staff data sheet
bench --site <site> execute medical_hrms.instructor_data_sheet.export_sheet
bench --site <site> execute medical_hrms.instructor_data_sheet.import_sheet --kwargs "{'file':
'/private/files/<name>.xlsx'}"
```

Documents in the repository: `README.md`, `DEVELOPER_SETUP.md`, `medical_hrms/jameah/README.md` and `VIEWS.md` (every Ministry column and its source), `project_tracking/` (Task Tracker, Test Cases, Checklist).